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Lamp with light source and extending reflection area

Abstract

Disclosed is a lamp including a light source that outputs light, a light output part that outputs a first light being any portion of the light output from the light source, and a reflection area that reflects any portion of the first light, the light output optical axis extends along a forward/rearward direction, and the reflection area extends to a rear side to be inclined to an upper side with respect to the light output optical axis, or extends in parallel to the light output optical axis.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority to Korean Patent Application No. 10-2023-0103049, filed in the Korean Intellectual Property Office on Aug. 7, 2023, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a lamp.

BACKGROUND

(3) Various types of vehicle lamps are mounted on vehicles depending on their functions. For example, a low beam lamp, a high beam lamp, and a daytime running light (DRL) lamp are mounted on a front side of a vehicle. Among them, the low beam lamp forms a light distribution pattern having a cutoff line shape at an upper portion thereof.

(4) In detail, the low beam lamp has a primary optical system that collects the light output from a light source, a shield that forms a cutoff shape, and a secondary optical system that focuses on the shield, and outputs the light to a front side to form a light distribution pattern. The shield is provided with a reflection area that may reflect any portion of the light that passed through the primary optical system.

(5) Any portion of the light reflected from the reflection area provided in the conventional lamp fails to reach the light output surface of the secondary optical system, and is completely reflected by the light output surface and is not output to a front side. Because the light that is not output to the front side cannot be used to form a light distribution pattern, a light output efficiency of the lamp is decreased.

SUMMARY

(6) The present disclosure has been made to solve the above-mentioned problems occurring in the prior art while advantages achieved by the prior art are maintained intact.

- (7) An aspect of the present disclosure provides a lamp that minimizes failure of light reflected from a reflection area to reach a light output surface whereby a light output efficiency is improved.
- (8) The technical problems to be solved by the present disclosure are not limited to the aforementioned problems, and any other technical problems not mentioned herein will be clearly understood from the following description by those skilled in the art to which the present disclosure pertains.
- (9) According to an aspect of the present disclosure, a lamp includes a light source that outputs light, a light output part that outputs a first light being any portion of the light output from the light source, and a reflection area that reflects any portion of the first light, the light output optical axis extends along a forward/rearward direction, and the reflection area extends to a rear side to be inclined to an upper side with respect to the light output optical axis, or extends in parallel to the light output optical axis.
- (10) Furthermore, the light output optical axis may extend in the forward/rearward direction, and the reflection area may extend to a rear side to be inclined to an upper side with respect to the light output optical axis.
- (11) Furthermore, an upward/downward height of a front end portion of the reflection area may be disposed to be the same as an upward/downward height of the light output optical axis, and a rear end portion of the reflection area may be disposed on an upper side of the light output optical axis.
- (12) Furthermore, an angle formed by or between the reflection area and the light output optical axis may be more than 0 degrees and not more than 5 degrees.
- (13) Furthermore, the lamp may further include a light input part, through which the light output from the light source passes, wherein the reflection area reflects the light passing through the light input part, and an angle formed by or between the reflection area and the light output optical axis may be the same as an angle formed by or between a light input optical axis being an optical axis of the light input part and the light output optical axis.
- (14) Furthermore, the lamp may further include a light shielding area that shields a second light being another portion of the light output from the light source such that the second light is prevented from being output from the light output part, and an extension area extending from a rear end portion of the reflection area to a lower side, and connected to a front end portion of the light shielding area, and an upward/downward height of the front end portion of the light shield area may be disposed to be the same as an upward/downward height of the light output optical axis.
- (15) Furthermore, the lamp may further include a light input part, through which the light output from the light source passes, and the light input part, the light shielding area, the extension area, the reflection area, and the light output part may be integrally formed.
- (16) Furthermore, the light source, the light shielding area, the extension area, the reflection area, and the light output part may be sequentially arranged along the forward/rearward direction.
- (17) Furthermore, the reflection area may extend in parallel to the light output optical axis, and an upward/downward height thereof may be disposed to be the same as an upward/downward height of the light output optical axis.
- (18) Furthermore, the reflection angle may be smaller than a reference angle being an angle formed by or between a reference line passing through an upper end of the light output part and a front end portion of the reflection area and the light output optical axis when the lamp is viewed in a leftward/rightward direction.
- (19) Furthermore, when any one arbitrary ray of the first light is a first ray, and a travel direction of the first ray before the first ray reaches the reflection area is a reflection introduction direction, and a travel direction of the first ray reflected in the reflection area is a reflection direction, a reflection angle being an angle formed by or between a light output optical axis being an optical axis of the light output part and an imaginary straight line extending in the reflection direction may be equal to or smaller than a reflection introduction angle being an angle formed by the light output optical axis and an imaginary straight line extending in the reflection introduction direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other objects, features and advantages of the present disclosure will be more apparent from the following detailed description taken in conjunction with the accompanying drawings:
- (2) FIG. 1 is a perspective view of a lamp according to a first embodiment of the present disclosure;
- (3) FIG. 2 is a longitudinal cross-sectional view of a light output part according to a first embodiment of the present disclosure;
- (4) FIG. 3 is a longitudinal cross-sectional view of a lamp according to a first embodiment of the present disclosure;
- (5) FIG. 4 is a view illustrating a light input part provided with an aspheric lens;
- (6) FIG. 5 is a view illustrating the light input part provided with a TIR lens;
- (7) FIG. 6 is a view illustrating a light source and a light input part arranged along an upward/downward direction;
- (8) FIG. 7 is a longitudinal cross-sectional view of a lamp according to a first modification of the present disclosure;
- (9) FIG. 8 is a longitudinal cross-sectional view of a lamp according to a second embodiment of the present disclosure;
- (10) FIG. 9 is a longitudinal cross-sectional view of a lamp according to a second modification of the present disclosure;
- (11) FIG. 10 is a longitudinal cross-sectional view of a first optical module according to a third embodiment of the present disclosure;
- (12) FIG. 11 is a longitudinal cross-sectional view of a second optical module according to a third embodiment of the present disclosure;
- (13) FIG. 12 is a longitudinal cross-sectional view of a lamp according to a fourth embodiment of the present disclosure; and
- (14) FIG. 13 is a transverse cross-sectional view of a lamp according to a fifth embodiment of the present disclosure.

DETAILED DESCRIPTION

(15) Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. In adding reference numerals to the components of the drawings, it is noted that the same components are denoted by the same reference numerals even when they are drawn in different drawings. Furthermore, in describing the embodiments of the present disclosure, when it is determined that a detailed description of related known configurations and functions may hinder understanding of the embodiments of the present disclosure, a detailed description thereof will be omitted.

(16) Furthermore, in describing the components of the embodiments of the present disclosure, terms, such as first, second, “A,” “B,” (a), and (b) may be used. The terms are simply for distinguishing the components, and the essence, the sequence, and the order of the corresponding components are not limited by the terms. When it is described that a certain component is “input to,” “passes through” or “output from” a second component, it should be understood that the component may be directly input to, passes through, or output from the second component, but a third component may be “input” or “pass,” or “output” between the components.

First Embodiment

(17) Hereinafter, a lamp **10a** according to a first embodiment of the present disclosure will be described with reference to FIGS. 1 to 6.

(18) Referring to FIG. 1, the lamp **10a** may be a vehicle lamp that is provided in a vehicle. For

example, the lamp **10a** may be a headlamp that is provided on both left and right sides of a front side of the vehicle. The lamp **10a** may form a low beam light distribution pattern to ensure a visibility of a driver. The low beam light distribution pattern may include a cutoff line, a hot zone, and a wide zone. The cutoff line may define an upper end of the low beam light distribution pattern. The hot zone may define a central portion of the low beam light distribution pattern. Furthermore, the wide zone may define a peripheral portion of the light distribution pattern, which surrounds the hot zone, and may have a luminous intensity that is lower than that of the hot zone. The lamp **10a** may include a light source **100a**, a light input part **200a**, a light output part **300a**, a shield part **400a**, and a guide part **500a**.

(19) The light source **100a** may output light. As an example, the light source **100a** may be a light emitting diode (LED). The light output from this light source **100a** may be spread around a light source optical axis **100x**. The light source optical axis **100x** may mean an optical axis of the light source **100a**. The light source optical axis **100x** may be defined as an imaginary straight line that passes through a center of the light source **100a** and extends along a travel direction of the output light. The light source optical axis **100x** may be formed to be inclined toward a front side and a lower side with respect to a forward/rearward direction “A.”

(20) The light emitted from the light source **100a** may be input to the light input part **200a**. The light that passed through the light input part **200a** may be input to the guide part **500a**. The light input to the guide part **500a** may travel toward a focus of the light output part **300a**. In other words, the light input to the guide part **500a** may be condensed at a focus of the light output part **300a**.

(21) The light input part **200a** may be disposed on a front side of the light source **100a**. Furthermore, as an example, the light input part **200a** may be spaced apart from the light source **100a**. Furthermore, a front end portion of the light input part **200a** may be attached to a rear end portion of the guide part **500a**. In other words, the light input part **200a** may be disposed between the light source **100a** and the guide part **500a** with respect to the forward/rearward direction “A.”

(22) A light input groove that is recessed to a front side may be formed in the light input part **200a**. The light output from the light source **100a** may pass through the light input groove, and then may reach a light input area that is an area that surrounds the light input groove. The light that reached the light input area may be refracted in the light input area.

(23) Referring back to FIG. 3, a light input optical axis **200x** that is an optical axis of the light input part **200a** may be formed in parallel to a light source optical axis **100x**. The light input optical axis **200x** may be defined as an imaginary straight line that passes through a center of the light input part **200a** and a focus of the light output part **300a**. Furthermore, the focus of the light output part **300a** may be disposed on an upper side of a front end portion of a reflection area **410a** that will be described later.

(24) The features of the light source **100a**, the light input part **200a**, the light output part **300a**, the shield part **400a**, and the guide part **500a** described in the specification may be understood as features that may satisfy geometric characteristics, in which the light input optical axis **200x** passes through a center of the light input part **200a** and a focus of the light output part **300a**.

(25) Furthermore, the geometric characteristics of the light source optical axis **100x**, the light input optical axis **200x**, and a light output optical axis **300x** in the specification may be satisfied by a combination of at least some of the configurations according to first to fifth embodiments in the specification, and the satisfaction is not limited to a combination of the configurations according to any one embodiment.

(26) In detail, any one of light sources **100a**, **100b**, **100c1**, **100c2**, **100d**, and **100e**, any of light input parts **200a**, **200a'**, **200a''**, **200a'''**, **200b**, **200c1**, **200c2**, **200d**, and **200e**, any one of light output parts **300a**, **300b**, **300c1**, **300c2**, **300d**, and **300e**, any one of shield parts **400a**, **400b**, **400c1**, **400c2**, and **400d**, and any one of guide parts **500a**, **500b**, **500c1**, **500c2**, **500d**, and **500e** that are components according to the first to fifth embodiments in the specification may be combined with each other, and only the components of any one embodiment are not combined in a limited way.

(27) The light input optical axis **200x** and the light output optical axis **300x** may be nonparallel to each other. The light output optical axis **300x** may mean an optical axis of the light output part **300a**. For example, the light input optical axis **200x** may be an imaginary straight line that extends to be inclined to a front side and a lower side with respect to the light output optical axis **300x**.
(28) In other words, when the lamp **10a** is viewed in the leftward/rightward direction, the light input optical axis **200x** and the light output optical axis **300x** may cross each other at one point. As a detailed example, the light input optical axis **200x** and the light output optical axis **300x** may cross each other on an upper side of a front end portion of the reflection area **410a**.

(29) Meanwhile, the idea of the present disclosure is not limited thereto, and according to the first modification of the present disclosure, the light input optical axis **200x** and a light source optical axis **100x-1** may be nonparallel to each other. Referring to FIG. 7, the light source optical axis **100x-1** may be formed in parallel to a forward/rearward direction "A." In other words, when the lamp **10a** is viewed in the leftward/rightward direction, the light source optical axis **100x-1** and the light input optical axis **200x** may cross each other at one point. Furthermore, the light source optical axis **100x-1** may be disposed in parallel to the light output optical axis **300x**.

(30) In this way, when the light source optical axis **100x-1** is formed in parallel to the forward/rearward direction "A," a heat sink (not illustrated), on which the light source **100a** is seated, may be oriented not to be inclined with respect to the forward/rearward direction "A." As an example, the heat sink may be disposed on a rear side of the light source **100a**.

(31) Furthermore, a front end portion of a reflection area **410b** may be disposed on a lower side of the focus of the light output part **300a**, or may be located at the same height as that of the focus.

(32) The light input part **200a** may include a light input central part **210a** and a light input outskirt part **220a**.

(33) The light input to the light input central part **210a** may be refracted in the light input central part **210**, and then may travel to the focus of the light output part **300a**. The light input central part **210a** may define a central portion of the light input part **200a**. The light input central part **210a** may be disposed to face the light source **100a**. The light input central part **210a** may have a shape that is convex to a rear side. The rear side of the light input central part **210a** may define a front side of the light input area.

(34) The light input to the light input outskirt part **220a** may be totally reflected in the light input outskirt part **220a**, and then may travel to the focus of the light output part **300a**. For example, the light may be input to the inner area of the light input outskirt part **220a**, and the light may be totally reflected in the outer area of the light input outskirt part **220a**. An inner area of the light input outskirt part **220a** may define an inner peripheral surface of the light input area. For example, the inner area of the light input outskirt part **220a** may extend from a rear side of the light input central part **210a** to a rear side. Furthermore, an outer area of the light input outskirt part **220a** may extend from a rear end portion of the inner area of the light input outskirt part **220a** to a front side.

(35) The light that passed through the inner area of the light input outskirt part **220a** may be refracted, and then may reach the outer area of the light input outskirt part **220a**. The light input outskirt part **220a** may define an outskirt part of the light input part **200a**. The light input outskirt part **220a** may have a shape that surrounds the light input central part **210a**.

(36) Meanwhile, the light input part **200a** according to the first embodiment of the present disclosure is not limited to the above-described contents, and referring to FIGS. 4 and 5, light input parts **200a'** and **200a''** according to embodiments that are different from the first embodiment may be provided as collimators. The collimator may form parallel light (for example, light that is parallel to the forward/rearward direction).

(37) The light input parts **200a'** and **200a''** may be disposed between the light source **100a** and the guide part **500a** with respect to the forward/rearward direction "A." Furthermore, the light input parts **200a'** and **200a''** may be disposed to be spaced apart from the guide part **500a**.

(38) For example, referring to FIG. 4, the light input part **200a'** may have a shape of an aspheric

lens. As another example, referring to FIG. 5, the light input part **200a''** may have a shape of a total internal reflection (TIR) lens. The TIR lens may be named 'a total reflection lens.'

(39) Meanwhile, the light input part **200a'** and **200a''** according to the embodiments that are different from the first embodiment of the present disclosure are not limited to the above-described contents, and referring to FIG. 6, a light input part **200a'''** according to the embodiments that are different from the first embodiment may be oriented along an upward/downward direction "H."

(40) The light input part **200a'''** and the light source **100a** may be arranged to be spaced apart from each other along the upward/downward direction "H." For example, the light input part **200a'''** may be disposed on a lower side of the light source **100a**. Furthermore, the light input part **200a'''** may be attached to an upper end of the guide part **500a**.

(41) The guide part **500a** according to the first and other embodiments may include an upward/downward guide area, a reflection guide area, and a forward/rearward guide area. The upward/downward guide area may extend from the light input part **200a'''** to a lower side. The reflection guide area may extend from a lower end of the upward/downward guide area to a front side. In the reflection guide area, the light that passed through the light input part **200a'''** may be reflected, and may travel toward the shield part **400a**. The forward/rearward guide area may extend from a front end portion of the reflection guide area to the light output part **300a** along the forward/rearward direction "A."

(42) Referring back to FIG. 2, the light that passed through the guide part **500a** may be output from the light output part **300a** to an outside of the lamp **10a**. The light output from the light output part **300a** may travel to a front side. The light output part **300a** may be disposed on a front side of the guide part **500a**. The light output part **300a** may have a shape that is convex to a front side. As an example, the light output part **300a** may be an aspheric lens. The light output part **300a** may include an upper light output area **310a** and a lower light output area **320a**.

(43) The upper light output area **310a** may define an upper portion of the light output part **300a**. For example, the upper light output area **310a** may mean an area of the light output part **300a**, which is located on an upper side of the light output optical axis **300x**. The light output from the upper light output area **310a** may be named 'an upper light'. A curvature of the upper light output area **310a** in the upward/downward direction "H" may be named a first curvature.

(44) The upper light output area **310a** may have a shape that is symmetrical in the leftward/rightward direction. Furthermore, the upper light output area **310a** may have a shape that is rotationally symmetrical with respect to the light output optical axis **300x**. For example, when a right area (an area that defines a right side of the upper light output area **310a**) of the upper light output area **310a** is rotated about the light output optical axis **300x** by a specific angle, the right area of the rotated upper light output area **310a** may completely overlap a left area of the upper light output area **310a** (an area that defines a left side of the upper light output area **310a**) that is not rotated.

(45) The lower light output area **320a** may define a lower portion of the light output part **300a**. For example, the lower light output area **320a** may mean an area of the light output part **300a**, which is located on a lower side of the light output optical axis **300x**. The light output from the lower light output area **320a** may be named 'a lower light'. A curvature of the lower light output area **320a** in the upward/downward direction "H" may be named a second curvature.

(46) The lower light output area **320a** may have a shape that is symmetrical in the leftward/rightward direction. Furthermore, the lower light output area **320a** may have a shape that is rotationally symmetrical with respect to the light output optical axis **300x**. For example, when a right area (an area that defines a right side of the lower light output area **320a**) of the lower light output area **320a** is rotated around the light output optical axis **300x** by a specific angle, a right area of the rotated lower light output area **320a** may completely overlap a left area of the lower light output area **310a** (an area that defines a left side of the lower light output area **320a**) that is not rotated. The first curvature and the second curvature may be different from each other. For

example, when the lights having the same wavelength are output from the upper light output area **310a** and the lower light output area **320a**, respectively, a degree of refraction may be different. As another example, an upper light output focus that is a focus of the upper light output area **310a** and a lower light output focus that is a focus of a lower light output area **320d** may be different from each other.

(47) The upper light output focus may be located on the light output optical axis **300x**.

Furthermore, the lower light output focus may be located on a rear side of the upper light output focus. For example, when parallel light that travels toward a rear side is input to the lower light output area **320a** from an outside, the parallel light input to the lower light output area **320d** may pass through an area located on a rear side of the upper light output focus.

(48) Furthermore, when the first curvature and the second curvature are different, as an example, a step may be formed in a connection area that is an area, in which the upper light output area **310a** and the lower light output area **320a** are connected to each other.

(49) As another example, when no step is formed in the connection area, the upper light output area **310a** has a shape that is symmetrical in the leftward/rightward direction and is rotationally symmetrical with respect to the light output optical axis **300x**, and the lower light output area **320d** may have a shape that is symmetrical in the leftward/rightward direction and is not rotationally symmetrical with respect to the light output optical axis **300x**. However, the description of the shapes of the upper light output area **310a** and the lower light output area **320a** corresponds to an exemplary description, and the idea of the present disclosure is not necessarily limited to the examples.

(50) The first curvature may be greater than the second curvature. In other words, a degree of bending of the upper light output area **310a** with respect to the upward/downward direction “H” may be greater than a degree of bending of the lower light output area **320a**. That is, a radius of curvature of the upper light output area **310a** in the upward/downward direction “H” may be smaller than a radius of curvature of the lower light output area **320a** in the upward/downward direction “H.”

(51) For example, with reference to the lights having the same wavelength, a degree of refraction of the light output from an upper reference point is refracted may be greater than a degree of refraction of the light output from a lower reference point. For example, the light output from the upper reference point may be refracted to be inclined toward a lower side with respect to a travel direction of the input light. Furthermore, the light output from the lower reference point may be refracted to be inclined toward an upper side with respect to the travel direction of the input light.

(52) The upper reference point may mean an arbitrary point of the upper light output area **310a**, which is spaced apart from the light output optical axis **300x** to an upper side, and the lower reference point may mean an arbitrary point of the lower light output area, which is spaced apart from the light output optical axis **300x** to a lower side. Furthermore, the upper reference point and lower reference point may be positioned at positions that are perpendicular to the upward/downward direction “H” and are symmetrical to each other with respect to the imaginary plane that passes through the light output optical axis **300x**. That is, a spacing distance between the upper reference point and the imaginary plane in the upward/downward direction “H” may be the same as a spacing distance between the lower reference point and the imaginary plane in the upward/downward direction “H.”

(53) Furthermore, an angle formed by or between the light input to the upper reference point and the light output optical axis **300x** may be the same as an angle formed by or between the light input to the lower reference point and the light output optical axis **300x**.

(54) Referring back to FIG. 2, among the lights output from the upper reference point, the light having a first wavelength, the light having the second wavelength, and the light having the third wavelength may be named a first upper wavelength light **Rr1**, a second upper wavelength light **Rb1**, and a third upper wavelength light **Rg1**, respectively.

- (55) Furthermore, among the lights output from the lower reference point, the light having the first wavelength, the light having the second wavelength, and the light having the third wavelength may be named a first lower wavelength light **Rr2**, a second lower wavelength light **Rb2**, and a third lower wavelength light **Rg2**, respectively.
- (56) For example, the light having the first wavelength may be observed as red by the user. For example, the first wavelength may be 630 nm or more and 780 nm or less.
- (57) Furthermore, as an example, the light having the second wavelength may be observed as blue by the user. The second wavelength may be shorter than the first wavelength. As an example, the second wavelength may be 380 nm or more and 420 nm or less.
- (58) Furthermore, as an example, the light having the third wavelength may be observed as green. The third wavelength may be shorter than the first wavelength and longer than the second wavelength. As an example, the third wavelength may be greater than 420 nm and less than 630 nm.
- (59) Furthermore, a first upper wavelength angle **Ar1** that is an angle formed by or between the first upper wavelength light **Rr1** and the light output optical axis **300x** may be smaller than a first lower wavelength angle **Ar2** that is an angle formed by or between the first lower wavelength light **Rr2** and the light output optical axis **300x**.
- (60) Furthermore, a second upper wavelength angle **Ab1** that is an angle formed by or between the second upper wavelength light **Rb1** and the light output optical axis **300x** may be larger than a second lower wavelength angle **Ab2** that is an angle formed by or between the second lower wavelength light **Rb2** and the light output optical axis **300x**.
- (61) Furthermore, a third upper wavelength angle that is an angle formed by or between the third upper wavelength light **Rg1** and the light output optical axis **300x** may be smaller than a third lower wavelength angle **Ag2** that is an angle formed by or between the third lower wavelength light **Rg2** and the light output optical axis **300x**. Furthermore, as an example, the third upper wavelength angle may be 0 degrees. In other words, the third upper wavelength light **Rg1** and the light output optical axis **300x** may be parallel to each other.
- (62) A portion of the upper light, which has the first wavelength, and a portion of the lower light, which has the second wavelength may form a cutoff line. For example, a portion of the upper light, which has the first wavelength, may be the first upper wavelength light **Rr1**, and a portion of the lower light, which has the second wavelength, may be the second lower wavelength light **Rb2**. As a more detailed example, the first upper wavelength light **Rr1** may be red light and the second lower wavelength light **Rb2** may be blue light, and the red light and the blue light may overlap each other at the cutoff line.
- (63) In this way, as light components having different wavelengths overlap each other at the cutoff line through the difference between the first curvature and the second curvature, a prominent appearance of light of a specific wavelength at the cutoff line due to chromatic aberration may be minimized. In other words, due to chromatic aberration of the light components having different wavelengths, the cutoff line that is observed as purple or blue to the user may be minimized whereby a light distribution pattern with a chromaticity that satisfies the laws may be implemented.
- (64) The shield part **400a** may prevent a portion of the light that passed through the light input part **200a** from traveling to a front side. As an example, the shield part **400a** may have a shape that is recessed into an upper side. Furthermore, the shield part **400a** may be disposed between the light input part **200a** and the light output part **300a** with respect to the forward/rearward direction “A.” The shield part **400a** may include a first area **410a**, a second area **420a**, and a third area **430a**.
- (65) The first area **410a** may reflect the first light that is any portion of the light that passed through the light input part **200a**. For example, the first light may be reflected from an upper surface of the reflection area **410a**. The first light reflected from the first area **410a** may be output from the upper light output area **310a**. The first area **410a** may be named ‘a reflection area **410a**’.
- (66) The second area **420a** may extend from a rear end portion of the first area **410a** to a rear side.

For example, the second area **420a** may extend obliquely a rear side to be inclined toward a lower side with respect to the light output optical axis **300x**. After passing through the light input part **200a**, the light that reached the second area **420a** is reflected to an upper side, and may be prevented from moving to a front side. The second area **420a** may be named 'a light shielding area **420a**'.

(67) The third area **430a** may extend from a front end portion of the first area **410a** to a lower side. For example, the third area **430a** may extend obliquely to a lower side to be inclined from a front end portion of the first area **410a** to a front side. When the third area **430a** is cut in the upward/downward direction "H," the third area **430a** may have a shape of a straight line. However, the present disclosure is not limited to the example, and when the third area **430a** is cut in the upward/downward direction "H," the third area may have a curved shape that is convex to a rear side.

(68) The guide part **500a** may guide travel of the light that passed through the light input part **200a** to the light output part **300a**. The guide part **500a** may be disposed between the light input part **200a** and the light output part **300a** with respect to the forward/rearward direction "A." For example, a rear end portion of the guide part **500a** may be connected to a front end of the light input part **200a**, and a front end portion of the guide part **500a** may be connected to the light output part **300a**. Furthermore, the light input part **200a**, the light output part **300a**, the shield part **400a**, and guide part **500a** may be integrally formed. The light input part **200a**, the light output part **300a**, the shield part **400a**, and the guide part **500a** may be one lens that is integrally formed.

Second Embodiment

(69) Hereinafter, a lamp **10b** according to the second embodiment of the present disclosure will be described with reference to FIGS. **8** and **9**. In a description of the lamp **10b** according to the second embodiment, a difference from the lamp **10a** according to the first embodiment in the present disclosure is mainly described.

(70) The lamp **10b** according to the second embodiment of the present disclosure may include a light source **100b**, a light input part **200b**, a light output part **300b**, a shield part **400b**, and a guide part **500b**. Meanwhile, for the contents regarding the light source **100b**, the light input part **200b**, the light output part **300b**, and the guide part **500b** according to the second embodiment of the present disclosure, the description of the light source **100a**, the light input part **200a**, the light output part **300a**, and the guide part **500a** according to the first embodiment of the present disclosure is used.

(71) Furthermore, the light input part **200b** may include a light input central part **210b** and a light input outskirts part **220b**. Furthermore, the light output part **300b** may include an upper light output area **310b** and a lower light output area **320b**. For the description of the light input central part **210b**, the light input outskirts part **220b**, the upper light output area **310b**, and the lower light output area **320b** according to the second embodiment of the present disclosure, the description of the light input central part **210a**, the light input outskirts part **220a**, the upper light output area **310a**, and the lower light output area **320a** according to the first embodiment of the present disclosure is used.

(72) The shield part **400b** may include a first area **410b**, a second area **420b**, and a third area **430b**. Meanwhile, for the contents regarding the second area **420b** and the third area **430b** according to the second embodiment of the present disclosure, the description of the second area **420a** and the third area **430a** according to the first embodiment of the present disclosure is used.

(73) Referring to FIG. **8**, the first area **410b** may be named 'a reflection area **410b**'. The reflection area **410b** may extend in parallel to the forward/rearward direction "A." Furthermore, the reflection area **410b** may overlap the light output optical axis **300x**. In other words, as an example, a height of the reflection area **410b** in the upward/downward direction "H" may be equal to or smaller than a height of the light output optical axis **300x**.

(74) For example, when a height of the reflection area **410b** in the upward/downward direction "H" is smaller than a height of the light output optical axis **300x**, the reflection area **410b** may be spaced

apart from the light output optical axis **300x** in the upward/downward direction “H.” As another example, when a height of the reflection area **410b** in the upward/downward direction is the same as a height of the light output optical axis **300x**, the reflection area **410b** may overlap the light output optical axis **300x**.

(75) The reflection area **410b** may be disposed on a lower side of the light source **100a**.

Furthermore, the reflection area **410b** may be disposed on a lower side of the center of the light input part **200a**.

(76) In this way, as the reflection area **410b** is parallel to the forward/rearward direction “A,” an angle formed by or between the light that reached the reflection area **410b** and the light output optical axis **300x** may be the same as an angle formed by or between the light reflected by the reflection area **410b** and the light output optical axis **300x**.

(77) For example, the reflection introduction angle and the reflection angle may be the same. The reflection introduction angle may be defined as an angle formed by or between the light output optical axis **300x** and an imaginary straight line that extends in a reflection introduction direction. Furthermore, the reflection angle may be defined as an angle formed by or between the light output optical axis **300x** and an imaginary straight line that extends in a reflection direction.

(78) The reflection introduction direction may be defined as a travel direction of the first ray before the first ray reaches the reflection area **410b**. The reflection direction may be defined as a travel direction of the first ray that is reflected in the reflection area **410b**. The first ray may mean that any portion of the light output from the light output part **300a**.

(79) Meanwhile, the idea of the present disclosure is not limited thereto, and referring to FIG. 9, the first area **410b1** according to a second modification of the present disclosure may include a reflection area **411b** and an extension area **412b**.

(80) Furthermore, the reflection area **411b** may extend in nonparallel to the light output optical axis **300x**. For example, the reflection area **411b** may extend obliquely to a rear side to be inclined to an upper side with respect to the light output optical axis **300x**.

(81) For example, when the lamp **10b** is viewed in the leftward/rightward direction, the reflection area **411b** and the light output optical axis **300x** may form a specific angle. An angle formed by or between the reflection area **411b** and the light output optical axis **300x** may be greater than 0 degrees and not more than 5 degrees. Furthermore, as an example, an angle formed by or between the light input optical axis **200x** and the light output optical axis **300x** may be the same as an angle formed by or between the reflection area **411b** and the output area **213b**.

(82) A height of a front end portion of the reflection area **411b** may be disposed to be the same as a height of the light output optical axis **300x** in the upward/downward direction “H.” Furthermore, a rear end portion of the reflection area **411b** may be disposed on an upper side of the light output optical axis **300x**. Due to the reflection area **411b**, the reflection angle may be smaller than the reflection introduction angle.

(83) Furthermore, the reflection angle may be smaller than a reference angle. The reference angle may be defined as an angle that is formed by a reference line that passes through an upper end of the light output part **300a** and a front end portion of the reflection area **411b**, and the light output optical axis **300x** when the lamp is viewed in the leftward/rightward direction. In this way, because the reflection angle is smaller than the reference angle, the reflection angle may be prevented from becoming excessively large. In this way, because the reflection angle is not formed to be excessively large, the reflected light reaches a rear side of the light output part **300a** whereby the light may be prevented from being totally reflected by the light output part **300a**. Furthermore, because the reflection angle is not formed to be excessively large, a light output efficiency of the lamp **10b** may be improved by maximizing an amount of the light that reaches the light output part **300a**.

(84) Furthermore, the extension area **412b** may extend from a rear end portion of the reflection area **411b** to a lower side. Furthermore, a lower end of the extension area **412b** may be connected to a

front end portion of the second area **420b**. In other words, the extension area **412b** may extend in the upward/downward direction “H” between the rear end portion of the reflection area and the front end portion of the second area **420a**. The second area **420b** may be named ‘a light shielding area **420b**’.

Third Embodiment

(85) Hereinafter, lamps **10c1** and **10c2** according to the third embodiment of the present disclosure will be described with reference to FIGS. **10** and **11**. The lamps **10c1** and **10c2** may include a first optical module **10c1** and a second optical module **10c2**.

(86) Referring to FIG. **10**, the first optical module **10c1** may form a wide zone of a low beam light distribution pattern. Meanwhile, the present disclosure is not limited to the example, and the first optical module **10c1** may form an entire low beam light distribution pattern.

(87) The first optical module **10c1** includes a first light source **100c1**, a first light input part **200c1**, a first light output part **300c1**, a first shield part **400c1**, and a first guide part **500c1**. For the description of the first light output part **300c1** and the first guide part **500c1** according to the third embodiment of the present disclosure, the description of the light output part **300a** and the guide part **500a** according to the first embodiment of the present disclosure is used.

(88) The first light output part **300c1** may include a first upper light output area **310c1** and a first lower light output area **320c1**. For the description of the first upper light output area **310c1** and the first lower light output area **320c1** according to the third embodiment of the present disclosure, the description of the upper light output area **310a** and the lower light output area **320a** according to the first embodiment is used.

(89) A height of a lower end of the first light input part **200c1** in the upward/downward direction “H” may be equal to or greater than a height of the first light output optical axis **300x 1** in the upward/downward direction. The first light output optical axis **300x** may mean an optical axis of the first light output part **300c1**.

(90) A center of the first light input part **200c1** may be located between an upper end of the light output part **300a** and the light output optical axis **300x** with respect to the upward/downward direction “H.” Furthermore, a width of the first light input part **200c1** in the upward/downward direction “H” may be smaller than a width of the first light output part **300c1** in the upward/downward direction “H.” For example, a width of the first light input part **200c1** in the upward/downward direction “H” may be equal to or greater than a width of the first upper light output area **310c1** in the upward/downward direction “H.” The first upper light output area **310a** may form an upper portion of the first light output part **300c1**. Furthermore, with respect to the forward/rearward direction “A,” the first light input part **200c1** may be disposed to face the first upper light output area **310c1**.

(91) The first light input optical axis **200x1** that is an optical axis of the first light input part **200c1** may be formed in parallel to the light source optical axis **100x1** that is an optical axis of the first light source **100c1**. Furthermore, the first light input optical axis **200x1** may be defined as an imaginary straight line that passes through a front end portion of the first reflection area **410c1** and a center of the first light input part **200c1**. Furthermore, the first light input optical axis **200x1** may extend obliquely to be inclined to a front side and a lower side with respect to the first light output optical axis **300x1**. The first light output optical axis **300x1** may mean an optical axis of the first light output part **300c1**. The first light output optical axis **300x1** may be parallel to the forward/rearward direction “A.” The first light input part **200c1** may include a first light input central part **210c1** and a first light input outskirt part **220c1**.

(92) A distance between the first light input central part **210a** and the first light source **100a** may be named a first distance **L1**. For example, the first distance **L1** may correspond to a depth of the first light input groove that is a light input groove of the first light input part **200c1**.

(93) The first light input groove may have a shape that is recessed to a front side on a rear side of the first light input part **200c1**. The first light input groove may be a groove that is surrounded by

an area, in which the light output from the first light source **100c1** of the first light input part **200c1** is input.

(94) The first shield part **400c1** may include a first area **410c1**, a second area **420c1**, and a third area **430c1**. For the description of the first area **410c1** according to the third embodiment of the present disclosure, the description of the first area **410b** according to the second embodiment of the present disclosure is used. The first area **410c1** may be named ‘a first reflection area **410c1**’.

(95) The second area **420c1** may be a surface that extends between a rear end portion of the first reflection area **410c1** and a lower end of the first light input part **200c1**. The second area **420c1** may be named ‘an extension surface **420c1**’. A “H” height of the first reflection area **411b** in the upward/downward direction may be equal to or greater than a height of a lower end of the first light input part **200c1** in the upward/downward direction “H.” In this way, when the first light input part **200c1** is disposed on an upper side of the reflection area **411b**, the light that passed through a lower portion of the first light input part **200c1** may not be shielded by the first shield part **400c1** and may be output from the first light output part **300c1**. In other words, because all the light that passed through the first light input part **200c1** is output from the first light output part **300c1**, an amount of the light output from the first optical module **10c1** is maximized. In this way, because the amount of the light output from the first optical module **10c1** is maximized, an output efficiency of the first optical module **10c1** may also be maximized.

(96) The extension surface **420c1** may extend obliquely to a rear side to be inclined to an upper side. An angle formed by or between the extension surface **420c1** and the first light output optical axis **300x1** may be smaller than an angle formed by or between the first light input optical axis **200x1** and the first light output optical axis **300x**. The extension surface **420c1** may be exposed to an outside of the first optical module **10c1**.

(97) The third area **430c1** may extend from a front end portion of the first reflection area **411b** to a lower side. For example, the third area **430c1** may extend perpendicular to the forward/rearward direction “A.” Furthermore, with respect to the forward/rearward direction “A,” the third area **430c1** may be disposed to face the first lower light output area **320c1**. Furthermore, a width of the third area **430c1** in the upward/downward direction “H” may be the same as a width of the first lower light output area **320a** in the upward/downward direction “H.”

(98) Referring to FIG. 11, the second optical module **10c2** may form a hot zone of a low beam light distribution pattern. Meanwhile, the present disclosure is not limited to the example, and the second optical module **10c2** may form an entire low beam light distribution pattern.

(99) The wide zone of the low beam light distribution pattern formed by the first optical module **10c1** and the hot zone of the low beam light distribution pattern formed by the second optical module **10c2** may overlap each other to form a low beam light distribution pattern.

(100) The second optical module **10c2** may include a second light source **100c2**, a second light input part **200c2**, a second light output part **300c2**, a second shield part **400c2**, and a second guide part **500c2**. For the description of each of the second light output part **300c2** and the second guide part **500a** according to the third embodiment of the present disclosure, the description of the light output part **300a** and the guide part **500a** according to the first embodiment of the present disclosure is used.

(101) The second light output part **300c2** may include a second upper light output area **310c2** and a second lower light output area **320c2**. For the description of each of the second upper light output area **310c2** and the second lower light output area **320c2** according to the third embodiment of the present disclosure, the description of the upper light output area **310a** and the lower light output area **320a** according to the first embodiment is used.

(102) The second light source optical axis **100x2** that is an optical axis of the second light source **100a** may extend in parallel to the forward/rearward direction “A.” The second light source optical axis **100x2** may be parallel to the second light input optical axis **200x2** that is an optical axis of the second light input part **200c2**. Furthermore, the second light source optical axis **100x2**, the second

light input optical axis **200x2**, and the second light output optical axis **300x2** may be parallel to each other. The second light output optical axis **300x2** may mean an optical axis of the second light output part **300c2**.

(103) A diameter of the second light input part **200c2** may be different from a diameter of the first light input part **200c1**. For example, the diameter of the first light input part **200c1** may be smaller than the diameter of the second light input part **200c2**. The diameter of the first light input part **200c1** may mean a width of the first light input part **200c1** in a direction that is perpendicular to the first light input optical axis **200x1**. Furthermore, the diameter of the second light input part **200c2** may mean a width of the second light input part **200c2** in a direction that is perpendicular to the second light input optical axis **200x**.

(104) The second light input part **200c2** may include a second light input central part **210c2** and a second light input outskirt part **220c2**. A spacing distance between the second light input central part **210a** and the second light source **100c2** may be named a second distance **L2**. For example, the second distance **L2** may correspond to a depth of the second light input groove that is a light input groove of the second light input part **200c2**.

(105) The second light input groove may have a shape that is recessed to a front side on a rear side of the second light input part **200c2**. The second light input groove may be a groove that is surrounded by an area, in which the light output from the second light source **100c2** of the second light input part **200c2** is input.

(106) Furthermore, a diameter of the first light input groove may be smaller than a diameter of the second light input groove. For example, the diameter of the first light input groove may mean a width of the first light input groove in a direction that is perpendicular to the first light input optical axis **200x1**. Furthermore, the diameter of the second light input groove may mean a width of the second light input groove in a direction that is perpendicular to the second light input optical axis **200x2**.

(107) Furthermore, a first brightness that is a maximum brightness (a maximum luminous intensity) of the light that passed through the first light input part **200c1** may be smaller than a second brightness that is a maximum brightness of the light that passed through the second light input part **200c2**. Furthermore, the maximum brightness of the first optical module **10c1** may be smaller than the maximum brightness of the second optical module **10c2**. Furthermore, an amount of the light output from the first optical module **10c1** may be greater than an amount of the light output from the second optical module **10c2**. In this way, the second optical module **10c2** has a structure that may implement a high brightness with a small amount of light.

(108) The brightness of the light that passed through the first light input part **200c1** may mean the number of, among the lights that passed through the first light input part **200c1**, rays that exist within a unit solid angle range in a direction of a first light source optical axis **100x1**. Furthermore, the brightness of the light that passed through the second light input part **200c2** may mean the number of, among the lights that passed through the second light input part **200c2**, rays that exist within a unit solid angle range in a direction of the second light source optical axis **100x2**.

(109) Furthermore, the second distance **L2** may be greater than the first distance **L1**. For example, as a difference between the second distance **L2** and the first distance **L1** increases, a difference between the second brightness and the first brightness may also increase. In other words, depending on a spacing distance between the light source and the light input outskirt part, the brightness of the light that passed through the light input part may be changed.

(110) The second shield part **400c2** may include a first area **410c2**, a second area **420c2**, a third area **430c2**, and a fourth area **440c2**. For the description of each of the first area **410c2** and the second area **420c2** according to the third embodiment of the present disclosure, the description of the first area **410a** and the second area **420a** according to the first embodiment of the present disclosure may be used.

(111) The third area **430c2** may extend from a front end portion of the second area **420c2** to a lower

side. For example, the third area **430c2** may extend in parallel to the upward/downward direction “H.”

(112) The fourth area **440c2** may extend obliquely to a rear side to be inclined from a lower end of the third area **430c2** to a lower side. A height of a lower end of the fourth area **440c2** in the upward/downward direction “H” may be the same as a height of a lower end of the second light output part **300c2** in the upward/downward direction “H.”

Fourth Embodiment

(113) Hereinafter, a lamp **10d** according to the fourth embodiment of the present disclosure will be described with reference to FIG. 12. In a description of the lamp **10d** according to the fourth embodiment, a difference from the lamps according to the first to third embodiments of the present disclosure will be mainly described.

(114) The lamp **10d** according to the fourth embodiment of the present disclosure may include a light source **100d**, a light input part **200d**, a light output part **300d**, a shield part **400d**, and a guide part **500d**. Meanwhile, for the description of the light source **100d**, the light input part **200d**, the light output part **300d**, and the guide part **500d** according to the fourth embodiment of the present disclosure, the description of the light source **100a**, the light input part **200a**, the light output part **300a**, and the guide part **500a** according to the first embodiment of the present disclosure is used.

(115) The light input part **200d** may include a light input central part **210d** and a light input outskirts part **220d**. Furthermore, the light output part **300d** may include an upper light output area **310d** and a lower light output area **320d**. Meanwhile, for the description of the light input central part **210d**, the light input outskirts part **220d**, the upper light output area **310d**, and the lower light output area **320d** according to the fourth embodiment of the present disclosure, the description of the light input central part **210a**, the light input outskirts part **220a**, the upper light output area **310a**, and the lower light output area **320a** according to the first embodiment of the present disclosure is used.

(116) The shield part **400d** may include a first area **410d**, a second area **420d**, and a third area **430d**. The first area **410d** may have a shape that protrudes to a lower side. However, the present disclosure is not limited to the example, and the first area **410d** may have a shape that is parallel to the light output optical axis **300x**. The first area **410d** may include a reflection area **411d** and an extension area **412d**.

(117) The reflection area **411d** may reflect any portion of the light that passed through the light input part **200d**. The reflection area **411d** may extend obliquely to a rear side to be inclined to a lower side with respect to the light output optical axis **300x**. For example, a height of the front end portion of the reflection area **411d** in the upward/downward direction “H” may be the same as a height of the light output optical axis **300x** in the upward/downward direction “H.” Furthermore, a rear end portion of the reflection area **411d** may be disposed on a lower side of the light output optical axis **300x**.

(118) The extension area **412d** may extend from a rear end portion of the reflection area **411d** to an upper side. For example, the extension area **412d** may extend obliquely to an upper side to be inclined to a rear side. The extension area **412d** may be connected to a front end portion of the second area **420d**. In other words, the extension area **412d** may extend obliquely between a front end portion of the second area **420d** and a rear end portion of the reflection area **411d**. A height of an upper end of the extension area **412d** in the upward/downward direction “H” may be the same as a height of the light output optical axis **300x** in the upward/downward direction “H.”

(119) The second area **420d** may prevent the light that passed through the light input part **200d** from traveling to a front side. The second area **420d** may be named ‘a light shielding area **420d**’. The light shielding area **420d** may be disposed on an upper side of a lower end of the light output part **300a**.

(120) The third area **430d** may define a front side of the shield part **400d**. The third area **430d** may be named ‘a front area **430d**’. Output reflected light may reach the front area **430d**. The output reflected light may mean, among the lights that reached the light output part **300d**, light that is

reflected (as an example, totally reflected) from the light output part **300d** and travels to a rear side.

(121) The front area **430d** may extend obliquely to a front side to be inclined to a lower side with respect to the light output optical axis **300x**. For example, the front area **430d** may extend obliquely to a front side from a front end portion of the reflection area **411d** to be inclined to a lower side.

(122) A height of an upper end of the front area **430d** in the upward/downward direction may be the same as a height of the light output optical axis **300x** in the upward/downward direction “H.” In other words, the upper end of the front area **430d** may cross the light output optical axis **300x**.

(123) Furthermore, when the lamp **10d** is viewed in the leftward/rightward direction, the front area **430d** may have a shape of a straight line. For example, when the lamp **10d** is viewed in the leftward/rightward direction, the front area **430d** may have a diagonal shape extending to a front side and a lower side. When the lamp **10d** is viewed in the leftward/rightward direction, the front area **430d** may have a shape that is inclined to a front side with respect to an upward/downward reference line Hx. The upward/downward reference line Hx may be defined as an imaginary straight line that passes through a front end portion of the reflection area **411d** and extends in the upward/downward direction “H.”

(124) In this way, because the front area **430d** has a shape that is inclined to a front side with respect to the upward/downward reference line Hx, the output reflected light that reached the front area **430d** may be totally reflected in the front area **430d** and input of the output reflected light to the light output part **300d** may be minimized. In this way, when the output reflected light is totally reflected in the front area **430d** and the input to the light output part **300d** is minimized, the output reflected light may be output from the light output part **300d** whereby glare to the user may be minimized.

(125) The output reflected light that reached the front area **430d** may be output from the front area **430d**. For example, the output reflected light may be refracted to an upper side when being output from the front area **430d**. The light refracted and output from the front area **430d** may be input to the first area **410d** (e.g., the reflection area **411d**). The light input to the first area **410d** may pass through a first guide area, which will be described later, and may be output from an upper end of the guide part **500d**.

(126) The guide part **500d** may include a first guide area and a second guide area. The first guide area may be disposed on an upper side of the reflection area **411d**. The first guide area may mean an area that is located on an upper side of the guide part **500d** with respect to the light output optical axis **300x**.

(127) The first guide area may extend from the upper light output area **310d** to a rear side. A width of the first guide area in the upward/downward direction “H” may be the same as a width of the upper light output area **310a** in the upward/downward direction “H.”

(128) The second guide area may mean an area that is located on a lower side of the guide part **500d** with respect to the light output optical axis **300x**. The second guide area may guide travel of the output reflected light to a rear side. In other words, the second guide area may guide the light reflected from the light output part **300d** to the front area **430d**. The second guide area may extend in the forward/rearward direction “A” between the front area **430d** and the lower light output area **320d**.

Fifth Embodiment

(129) Hereinafter, a lamp **10e** according to the fifth embodiment of the present disclosure will be described with reference to FIG. 13. In a description of the lamp **10e** according to the fifth embodiment, a difference from the lamps according to the first to fourth embodiments of the present disclosure are mainly described.

(130) The lamp **10e** according to the fifth embodiment of the present disclosure may include a light source **100e**, a light input part **200e**, a light output part **300e**, a shield part (not illustrated), and a guide part **500e**. For the description of the light source **100e**, the light input part **200e**, the light output part **300e**, and the shield part (not illustrated) according to the fifth embodiment of the

present disclosure, the description of the light source **100a**, the light input part **200a**, the light output part **300a**, and the shield part **400a** according to the first embodiment of the present disclosure is used.

(131) The light input part **200e** may include a light input central part **210e** and a light input outskirts part **220c**. Furthermore, the light output part **300d** may include an upper light output area and a lower light output area. Meanwhile, for the description of the light input central part **210e**, the light input outskirts part **220e**, the upper light output area, and the lower light output area according to the fifth embodiment of the present disclosure, the description of the light input central part **210a**, the light input outskirts part **220a**, the upper light output area **310a**, and the lower light output area **320d** according to the first embodiment of the present disclosure is used.

(132) The guide part **500c** may be divided into a plurality of areas with respect to the forward/rearward direction “A.” In at least some of the plurality of areas, a width of the first area in the leftward/rightward direction and a width of the second area in the leftward/rightward direction may be different from each other. For example, when the first area is located on a front side of the second area, a width of the first area in the leftward/rightward direction may be greater than a width of the second area in the leftward/rightward direction. In other words, in at least some of the plurality of areas, a width of the area relatively located on a front side in the leftward/rightward direction may be greater than a width of the area relatively located on a rear side in the leftward/rightward direction.

(133) For example, the guide part **500e** may have a shape, in which a width thereof in the leftward/rightward direction becomes smaller toward a rear side. On the left side and the right side of the guide part **500e**, inclined surfaces **500e1** that extend obliquely to a rear side to be inclined to an inside of the lamp **10e**, respectively, may be formed. The inclined surfaces **500e1** may form a specific angle “T” with a forward/rearward reference line Ax. The forward/rearward reference line Ax may mean an imaginary straight line that extends in the forward/rearward direction “A.”

(134) A width of the front end portion of the guide part **500e** in the leftward/rightward direction may be the same as a width of the light output part **300e** in the leftward/rightward direction. Furthermore, a width of the rear end portion of the guide part **500e** in the leftward/rightward direction may be equal to or greater than a width of the light input part **200e** in the leftward/rightward direction.

(135) In an inclined surface **500e1**, a portion of the light totally reflected by the light input outskirts part **220c** may be reflected. For example, an angle formed by or between the inclined ray and the forward/rearward reference line Ax before the inclined ray reaches the inclined surface **500e1** may be greater than an angle formed by or between the inclined ray reflected by the inclined surface **500e1** and the forward/rearward reference line Ax. The inclined ray may mean one arbitrary ray of the light that is totally reflected by the light input outskirts part **220e**.

(136) When the inclined ray is reflected by the inclined surface **500e1**, a degree of inclination of the inclined ray with respect to the forward/rearward reference line Ax may be reduced. In other words, when the inclined ray is reflected by the inclined surface **500e1**, a degree of bending may be reduced. Accordingly, the inclined ray reflected by the inclined surface **500e1** may travel to be close to parallel to the forward/rearward direction “A.” In this way, when an inclination of the inclined ray with respect to the forward/rearward reference line Ax is reduced by the inclined surface **500e1**, an amount of the light output from the light output part **300e** may be maximized.

(137) The inclined surface formed on a left side of the guide part **500e** may be named ‘a left inclined surface’ or ‘a first inclined surface’. The first inclined surface may have a shape that extends to a rear side to be inclined to a right side. Furthermore, the first inclined surface may reflect a portion of the light totally reflected by the light input outskirts part **220e** to a right side. For example, the first inclined surface may reflect a portion of the light totally reflected from a right side of the light input outskirts part **220e** to a right side. After being reflected by the first inclined surface, the light output from the light output part **300c** may form a right side of the wide zone.

(138) Furthermore, the inclined surface formed on the right side of the guide part **500e** may be named ‘a right inclined surface’ or ‘a second inclined surface’. The second inclined surface may have a shape that extends to a rear side to be inclined to a left side. Furthermore, the second inclined surface may reflect a portion of the light totally reflected by the light input outskirts part **220e** to a left side. For example, the second inclined surface may reflect a portion of the light totally reflected from a left side of the light input outskirts part **220e** to a left side. After being reflected from the second inclined surface, the light output from the light output part **300e** may form a left side of the wide zone.

(139) Furthermore, the left inclined surface and right inclined surface may have shapes that are symmetrical to each other with respect to a reference plane. The reference plane may be defined as an imaginary plane that is perpendicular to the leftward/rightward direction and passes through a center of the lamp **10c** (as an example, a center of the guide part **500c**).

(140) The lamp according to the present disclosure minimizes failure of light reflected from the reflection area to reach the light output surface whereby a light output efficiency is improved.

(141) Furthermore, in describing the components of the embodiments of the present disclosure, terms, such as first, second, “A,” “B,” (a), and (b) may be used. The terms are simply for distinguishing the components, and the essence, the sequence, and the order of the corresponding components are not limited by the terms. Unless defined differently, all the terms including technical or scientific terms have the same meanings as those generally understood by an ordinary person in the art, to which the present disclosure pertains. The terms, such as the terms defined in dictionaries, which are generally used, should be construed to coincide with the context meanings of the related technologies, and are not construed as ideal or excessively formal meanings unless explicitly defined in the present disclosure.

(142) The above description is a simple exemplary description of the technical spirits of the present disclosure, and an ordinary person in the art, to which the present disclosure pertains, may make various corrections and modifications without departing from the essential characteristics of the present disclosure. Therefore, the embodiments disclosed in the present disclosure are not for limiting the technical spirits of the present disclosure but for describing them, and the scope of the technical spirits of the present disclosure is not limited by the embodiments. The protection scope of the present disclosure should be construed by the following claims, and all the technical spirits in the equivalent range should be construed as being included in the scope of the present disclosure.

Claims

1. A lamp comprising: a light source disposed at a rear portion of the lamp and configured to output light; a light output part disposed at a front portion of the lamp and configured to output first light, which is a first portion of the light output from the light source; and a reflection area located between the light source and the light output part and configured to reflect a portion of the first light, wherein a light output optical axis, which is an optical axis of the light output part, extends in a forward/rearward direction of the lamp, wherein the reflection area extends toward the rear portion of the lamp and is inclined with respect to the light output optical axis, and wherein the reflection area includes: a front end portion disposed at a same height as the light output optical axis, and a rear end portion disposed above the light output optical axis.

2. The lamp of claim 1, wherein an angle between the reflection area and the light output optical axis is more than 0 degree and not more than 5 degrees.

3. The lamp of claim 1, further comprising a light input part, through which the light output from the light source passes, wherein the reflection area is configured to reflect the light passing through the light input part, and wherein an angle between the reflection area and the light output optical axis is same as that between a light input optical axis, which is an optical axis of the light input part, and the light output optical axis.

4. The lamp of claim 1, further comprising: a light shielding area configured to shield a second light, which is a second portion of the light output from the light source, such that the second light is prevented from being output from the light output part; and an extension area extending from a rear end portion of the reflection area to a lower side of the lamp and connected to a front end portion of the light shielding area, wherein the front end portion of the light shielding area has a same height as the light output optical axis.

5. The lamp of claim 4, further comprising a light input part, through which the light output from the light source passes, wherein the light input part, the light shielding area, the extension area, the reflection area, and the light output part are integrally formed.

6. The lamp of claim 4, wherein the light source, the light shielding area, the extension area, the reflection area, and the light output part are sequentially arranged along the forward/rearward direction.

7. The lamp of claim 1, wherein: the first light includes a first ray, which travels in a reflection introduction direction before reaching the reflection area and then travels in a reflection direction after being reflected by the reflection area, and a reflection angle between (1) a light output optical axis, which is an optical axis of the light output part, and (2) a first imaginary straight line extending in the reflection direction is equal to or smaller than a reflection introduction angle between (1) the light output optical axis and (2) a second imaginary straight line extending in the reflection introduction direction.
